F79SP Session 14: Further theories on stationary distributions and limiting distributions (1)

Recall that we have the following nice result.
Result: Suppose that a THMC is with **finite state space S being a single closed class, and is aperiodic**.

- The existence and uniqueness of stationary distribution is guaranteed.
- In addition, it is always the associated limiting distribution of any initial distribution.

We will study on how the relaxation of the three properties: finite state space, irreducible and aperiodic will affect our result on stationary and limiting distribution.

1 Irreducible, aperiodic Markov chains

First, **Result** above can be reformulated as follows.

Theorem 1.1. *Suppose that a THMC is such that S* ***is a single closed class and is aperiodic****, then:*

1. *There exists a vector (not necessarily a valid distribution) $\boldsymbol{\omega} = (\omega_j)_{j\in S}$ such that for all $i, j \in S$, $\lim_{n\to\infty} p_{ij}^n = \omega_j$ (independent of i).*
2. *In addition, if S* ***is finite****, then $\boldsymbol{\omega}$ above is the unique stationary distribution.*

Proof. We will revisit this! □

We have the following remark for (1.) in Theorem 1.1.

Remark 1.1. *In other words,* $\lim_{n\to\infty} P^n = \begin{pmatrix} \boldsymbol{\omega} \\ \boldsymbol{\omega} \\ \vdots \end{pmatrix}$. *For finite S, for any initial distribution $\boldsymbol{\mu} = (\mu_j)_{j\in S}$,* $\lim_{n\to\infty} \boldsymbol{\mu} P^n = \boldsymbol{\mu} \lim_{n\to\infty} P^n = \sum_{i\in S} \mu_i \boldsymbol{\omega} = (\sum_{i\in S} \mu_i)\boldsymbol{\omega} = 1\boldsymbol{\omega} = \boldsymbol{\omega}$. *Every $\boldsymbol{\mu}$ converges to the same vector $\boldsymbol{\omega}$.*

Example 1.1. *Revisit Tutorial 3 Q3.*

- *Notice that the MC with transition matrix* $P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ 1 & 0 \end{pmatrix}$ *is irreducible and aperiodic.*
- *We have shown that* $\lim_{n\to\infty} P^n = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{pmatrix}$ *(as guaranteed by (1.) of Theorem 1.1, $\boldsymbol{\omega} = (\frac{2}{3}, \frac{1}{3})$).*
- *We validated that for two initial distributions $\boldsymbol{\mu}$, $\lim_{n\to\infty} \boldsymbol{\mu} P^n = \boldsymbol{\omega}$. In addition, this will hold for any initial distribution as mentioned in Remark 1.1.*
- *You may check that $\boldsymbol{\omega} = (\frac{2}{3}, \frac{1}{3})$ is a distribution, is stationary, and in addition, is the unique stationary distribution (as guaranteed by (2.) of Theorem 1.1).*

Infinite state space, irreducible and aperiodic

Either one of the following will hold:

1. The conclusion of **Result** holds (same as finite case).
2. There is no stationary distribution. The $\boldsymbol{\omega}$ mentioned in (1.) of Theorem 1.1 is $\boldsymbol{\omega} = (0, 0, \dots)$, not a valid distribution.

2 Finite state space

What can we say about the stationary and limiting distribution if the only underlying assumption of our MC is of **finite state space**?

Result 2.1. *For THMC with **finite state space**, we have the following:*

1. *A stationary distribution always exists, but it is not necessarily unique.*
2. *A limiting distribution may or may not exist (depending on initial distribution), and if exists, it is not necessarily unique.*
3. *Any limiting distribution (if exists) is stationary.*
4. *Any stationary distribution (if exists) is a limiting distribution for some initial distribution.*

Remarks: We make some remarks on each point in Result 2.1

1. We will see examples of non-unique stationary distributions in later discussion.
2. We will see examples of initial distribution with no limiting distribution exist in later discussion.
3. If there exists $\boldsymbol{\mu}$ such that $\lim_{n\to\infty} \boldsymbol{\mu}P^n = \boldsymbol{\omega}$ forms a distribution, then $\boldsymbol{\omega}P = \boldsymbol{\omega}$.
   Idea: If the sequence $(\boldsymbol{v}_n) = \boldsymbol{\mu}, \boldsymbol{\mu}P, \boldsymbol{\mu}P^2, \dots$ converges to $\boldsymbol{\omega}$, then for large n, $\boldsymbol{v}_n \approx \boldsymbol{v}_{n+1} = \boldsymbol{v}_n P$.
4. Suppose that $\boldsymbol{\pi}P = \boldsymbol{\pi}$. Note that by setting initial distribution $\boldsymbol{\mu} = \boldsymbol{\pi}$, we have $\lim_{n\to\infty} \boldsymbol{\mu}P^n = \boldsymbol{\pi}$.

2.1 Case 1: MC <u>has</u> exactly one closed class and it is aperiodic

The main ideas are as follows:

- Let $S = C \cup T$, C is the closed class, T is a finite union of all open classes.
- C itself forms a MC (of single closed class) with transition matrix $\tilde{P}$ being a reduction matrix of P.
- Note that a stationary distribution $\boldsymbol{\pi} = (\pi_i)_{i\in S}$ must satisfy $\pi_i = 0$ for every $i \in T$, $\sum_{i\in C} \pi_i = 1$, and $\pi_i > 0$ for every $i \in C$.
- The remaining π_i with $i \in C$ can be found by solving $\tilde{\boldsymbol{\pi}}\tilde{P} = \tilde{\boldsymbol{\pi}}$.
- This stationary distribution $\boldsymbol{\pi}$ is unique, and will be the limiting distribution of any initial distribution.

Example 2.1. $P = \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} & 0 \\ \frac{1}{4} & 0 & \frac{3}{4} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$.

2.2 Case 2: S is finite, irreducible and MC is periodic

- Let $d \geq 2$ be its period. We have a partition $S = S_1 \cup S_2 \cup \cdots \cup S_d$ such that $\mathbf{P}(X_1 \in S_{k+1} | X_0 \in S_k) = 1$.
- As S is finite and irreducible, stationary distribution exists and is unique.
- The stationary distribution $\boldsymbol{\pi} = (\pi_i)_{i \in S}$ must be of same proportion in each S_i, hence $\sum_{i \in S_1} \pi_i = \sum_{i \in S_2} \pi_i = \cdots = \sum_{i \in S_d} \pi_i = \frac{1}{d}$.
- We have an initial distribution $\boldsymbol{\mu} = (\mu_i)_{i \in S}$ converges to a limiting distribution **iff** $\sum_{i \in S_1} \mu_i = \sum_{i \in S_2} \mu_i = \cdots = \sum_{i \in S_d} \mu_i = \frac{1}{d}$. Since limiting distribution must be stationary, such limiting distribution, if exists, must be the $\boldsymbol{\pi}$ above.

Example 2.2. *Consider a MC with transition matrix* $P = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{2}{3} & 0 & 0 \\ \frac{4}{5} & 0 & \frac{1}{5} & 0 & 0 \end{pmatrix}$.

1. *Determine its stationary distribution.*
2. *Provide your justification on the limiting distribution given that the initial distribution is* $\mu = (\frac{1}{6}, \frac{1}{3}, \frac{1}{6}, \frac{1}{4}, \frac{1}{2})$.
3. *Provide your justification on the limiting distribution given that the initial distribution is* $\mu = (1, 0, 0, 0, 0)$.